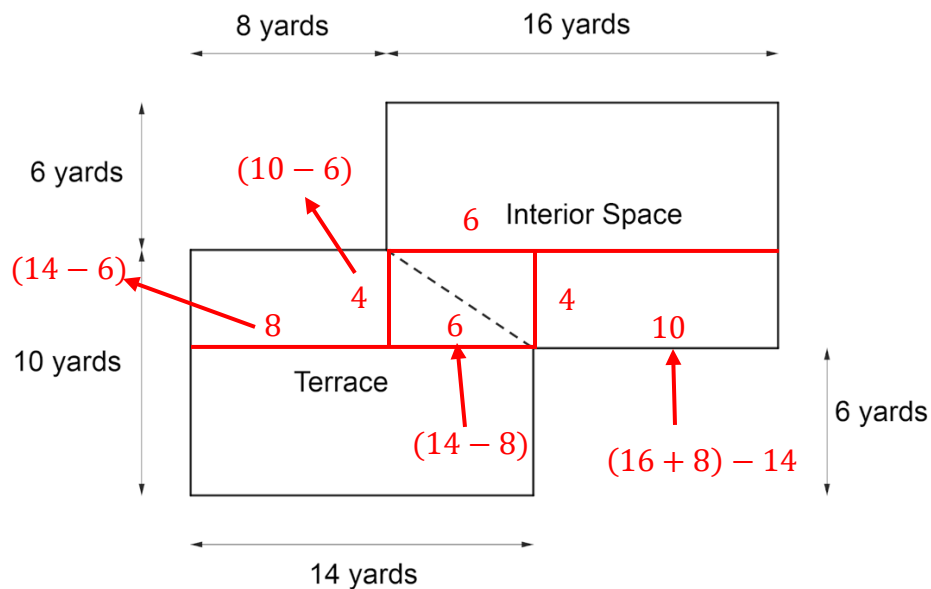


Grade 6 Accelerated
Unit 7
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

A local restaurant consists of both interior space and an outdoor terrace.



- Decompose the floorplan into triangles and quadrilaterals.
Answers may vary

- Find the area of the terrace.

$$(14 \cdot 6) + (8 \cdot 4) \rightarrow \left(\frac{1}{2} \cdot 6 \cdot 4\right)$$

$$84 + 32 + 12 = 128 \text{ yd}^2$$

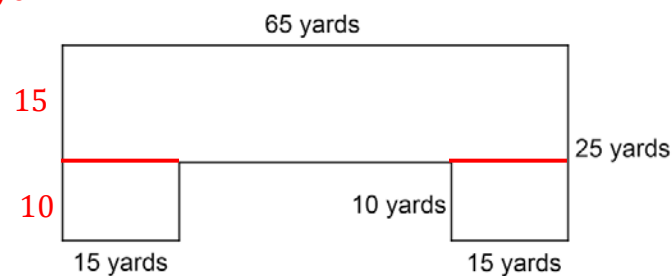
- Find the area of the interior space.

$$(6 \cdot 16) + (10 \cdot 4) + \left(\frac{1}{2} \cdot 6 \cdot 4\right)$$

$$96 + 40 + 12 = 148 \text{ yd}^2$$

- Find the area of the entire restaurant.

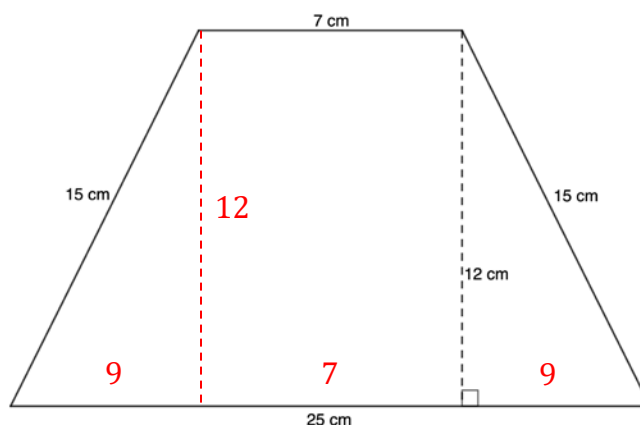
$$128 + 148 = 276 \text{ yd}^2$$



5. Decompose the image into quadrilaterals.

6. Find the area of the composite figure.

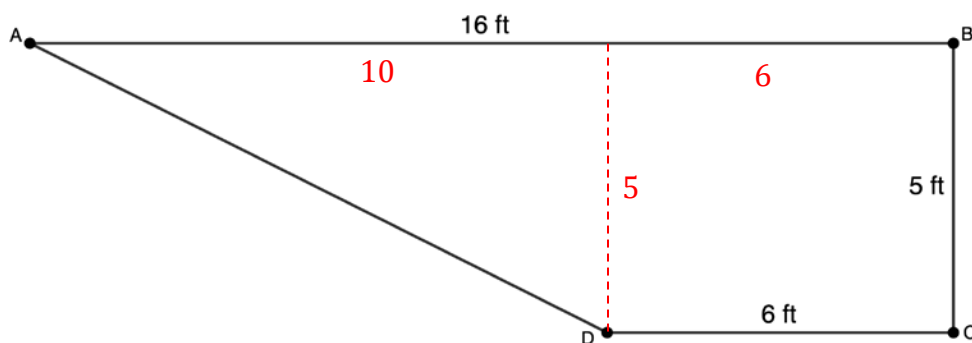
$$(10 \cdot 15) + (10 \cdot 15) + (15 \cdot 65) = 150 + 150 + 975 = 1275 \text{ yd}^2$$



$$25 - 7 = \frac{18}{2} = 9$$

7. Decompose the image into triangles and quadrilaterals.

8. Find the area of the image. $2 \left(\frac{1}{2} \cdot 9 \cdot 12 \right) + (7 \cdot 12) = 192 \text{ cm}^2$



9. Decompose the image into triangles and quadrilaterals.

10. Find the area of the composite figure.

$$\left(\frac{1}{2} \cdot 10 \cdot 5 \right) + (6 \cdot 5) = 25 + 30 = 55 \text{ ft}^2$$